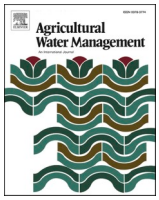
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Mechanisms and regulatory pathways for increasing water and nitrogen use efficiencies by crop diversification: A review

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ABSTRACT

Crop diversification plays a vital role in sustainable agriculture by enhancing ecosystem functions and improving resource efficiency. It is a key strategy to ensure food security while optimizing water and nitrogen use, addressing the challenges posed by limited resources, and supporting environmental preservation. This review synthesizes current knowledge regarding the mechanisms and regulatory pathways through which diversified cropping systems enhance water productivity (WP) and agronomic efficiency of nitrogen (AE_N), and partial factor productivity of nitrogen (PFP_N). It also identifies existing challenges and proposes priority directions for future research. This analysis focuses on studies of diversified cropping systems, such as intercropping, crop rotation, and multiple cropping, that enhance WP, AE_N, and PFP_N. We elucidate the underlying mechanisms driving these improvements by incorporating recent findings. Diversified cropping systems enhance WP, AE_N, and PFP_N by reducing water and nitrogen losses, improving soil physicochemical properties, regulating interspecific interactions, enhancing crop physiological performance, and stimulating beneficial microbial activity. These benefits are optimized through strategic crop combinations, integrated water–nitrogen management, improved tillage and mulching practices, and precision monitoring and decision-support systems. Despite these advantages, challenges remain in crop variety selection, mechanization, and the comprehensive evaluation of socio-economic and ecological outcomes. Future research should emphasize system adaptability to climate variability, the molecular mechanisms underlying water and nitrogen use, water–carbon–nitrogen coupling, artificial intelligence (AI)-assisted system design, and implementation-oriented strategies. The efficacy of diversified systems and associated management practices is strongly influenced by local climatic conditions, soil properties, and socio-economic constraints. This review highlights the need for an integrated framework combining agronomic, physiological, and molecular perspectives to better understand and regulate how crop diversification enhances WP, AE_N, and PFP_N. The findings provide a theoretical and technological foundation for promoting diversified, resource-efficient, and climate-resilient cropping systems, particularly in irrigated regions of Northwest China and similar climatic environments.

1. Introduction

The global food security in the 21st century faces unprecedented challenges driven by rapid population growth, escalating climate change impacts, and degradation of natural resources (Beltran-Peña et al., 2020; Herrero et al., 2020). Modern agriculture is facing the dual responsibility of meeting growing global food demands while also curbing environmental degradation. In this context, water scarcity and nitrogen (N) pollution represents two of the most critical constraints to sustainable intensification (Deng et al., 2025). The effective use of water and N in crop production systems becomes the key point for sustainable crop production. Water scarcity currently impacts more than 40% of the global population, and this value is projected to increase under changing climate conditions, especially since water productivity (WP) in crop production systems remains relatively low (Deng et al., 2025; Jiang et al., 2025). At the same time, the partial factor productivity of nitrogen (PFP_N) in crop production remains very low, as crops typically utilize only 30–50% of the applied N fertilizer; the remainder is lost to the

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environment through leaching, volatilization, and denitrification (Zhang et al., 2015; Liu et al., 2024b). Not only does this indicate substantial economic loss for the crop grower but also environmental concerns such as groundwater contamination and the eutrophication and the release of the greenhouse gas N_2O (Wang et al., 2022a; Liu et al., 2024b). Thus, improving water productivity and N use efficiency provides synergistic benefits for balancing agricultural productivity, environmental protection, and social welfare.Tables 1 and 2

Water availability inherently regulates N dynamics in soil−plant systems, while N mobility and uptake are predominantly dependent on available soil water (Wang et al., 2014; Guo et al., 2022). Correspondingly, N status affects WP of crops via root growth, stomatal conductance, and photosynthetic efficiency (Ye et al., 2022; Li et al., 2025b). The close interaction indicates that to improve WP, AE_N, and PFP_N should be addressed jointly rather than individually. Traditional methods for improving WP, AE_N, and PFP_N include the optimization of agronomic practices such as irrigation and N scheduling (Guo et al., 2022; Xu et al., 2023), the use of N from the controlled-release and organic N fertilizers (Guo et al., 2022), and appropriate tillage and mulching practices (Masvaya et al., 2017; Yin et al., 2020a). Despite some beneficial effects from these approaches, the efficiency gains remain limited due to high implementation costs, technological barriers for the small-scale farmers, and the complex interactions between the N management practices and local environmental conditions.

Crop diversification has emerged as a promising complementary or alternative approach for simultaneously enhancing WP, AE_N, and PFP_N (Yin et al., 2024; Yang et al., 2025b; Yi et al., 2025). Defined as the integration of multiple crop species within spatial or temporal production systems, diversification enhances resource-use efficiency through complementary resource capture in space and time, improved soil structure and water retention, increased microbial activity and nutrient cycling, and reduced pest and disease pressures, ultimately leading to improved soil health (He et al., 2024; Yang et al., 2024). Legume-based rotations and intercropping systems contribute substantial N inputs via biological fixation while enhancing soil water-holding capacity (Yang et al., 2024; Li et al., 2025a). Similarly, intercropping deep- and shallow-rooted species enhances access to water and nutrients from different soil horizons, outperforming monocultures in this regards (Chai et al., 2021; Zhang et al., 2024a). Earlier findings suggest that nitrogen use efficiency (AE_N and PFP_N) in common, particularly small-grain cereals, is generally higher compared with legume species, which exhibit high N acquisition potential due to N-fixing activity via symbiosis with nodulating Rhizobia (Adams et al., 2017; Qiao et al., 2024). Thus, crop species exhibit different strategies for the acquisition and utilization of water and N, when used together in diversified systems, these complementary traits can result in more efficient resource use. Recent molecular and genomic advances have begun to elucidate the genetic and regulatory mechanisms underlying these processes, including the roles of N transporters, aquaporins, and key N-metabolizing enzymes such as nitrate reductase and glutamine synthetase (Dong et al., 2024d; Qiao et al., 2024). Despite growing recognition of the potential benefits of crop diversification, our understanding of the mechanisms and regulatory pathways through which it enhances WP, AE_N, and PFP_N remains fragmented. The bulk of the studies has concentrated on simplified crops or limited systems, which may overlook emergent properties arising from species interactions in diversified systems (Yin et al., 2020a; Demie et al., 2025). Moreover, the direct effects of the diversified practices (e.g., crop rotation, intercropping, multiple cropping) on improving WP, AE_N, and PFP_N are not well quantified, particularly with fluctuating environmental conditions. Little is also known about the effect of crop diversification on complex feedback loops among soil microbial communities, nutrient cycles, and water-nitrogen interactions, which ultimately determine system efficiency at boarder scales.

This review aims to synthesize current knowledge on the mechanisms and regulatory processes through which crop diversification

Table 1
Typical examples of improved water productivity (WP) with crop diversification in countries with highly diversified cropping systems.

Country	Cropping system	Crop configuration	References
China	Intercropping	Proso millet||mung bean, maize||soybean, maize||wheat, maize||alfalfa, maize||peanut intercropping	(Gong et al., 2021; Zhang et al., 2022c; Fan et al., 2023; Hou et al., 2024; Sun et al., 2025)
	Rotation	Maize→wheat, maize→rice, cabbage→oat, rice→wheat, spring millet→wheat-maize rotation	(Hu and Lei, 2021; Huang et al., 2022; Yu et al., 2023; Zhao et al., 2024a; Yang et al., 2025b)
	Multiple cropping	Winter wheat—summer maize, peanut—wheat, wheat—green manure; wheat—soybean multiple cropping	(Yang et al., 2022; Zhang et al., 2022a; Wang et al., 2023, 2024a; Dong et al., 2025)
India	Intercropping	Rice||fodder, maize||blackgram, sorghum||cowpea, grass||groundnut, chickpea||quinoa intercropping	(Bastia et al., 2021; Varatharajan et al., 2022; Tanwar et al., 2023; Lal et al., 2024; Pradhan et al., 2025)
	Rotation	Maize→wheat, cluster bean→wheat; mung bean→wheat; rice→wheat rotation	(Jat et al., 2013; Rathore et al., 2014; Abrol et al., 2024)
	Multiple cropping	Maize—wheat, rice—wheat multiple cropping	(Sandhu et al., 2019; Jyolsna et al., 2024)
USA	Intercropping	Peanut||watermelon, maize||soybean intercropping	(Franco et al., 2018; Pelech et al., 2021)
	Rotation	Maize→soybean, winter wheat→spring barley/carinata, millet→cowpea rotation	(Hussain et al., 2019; Williams et al., 2020; Scott et al., 2021; Darapuneni et al., 2022)
	Multiple cropping	Camelina—soybean, soybean—wheat, winter wheat—summer green manure multiple cropping	(Gesch and Johnson, 2015; Baath et al., 2021; Singh et al., 2023)
Brazil	Intercropping	Maize||cowpea, cactus||millet, cactus||sorghum intercropping	(Júnior et al., 2023; Silva et al., 2023, 2024; Salvador et al., 2025)
	Rotation	Sorghum→sweet maize, rice→soybean rotation	(Santos et al., 2022; Cunha et al., 2025)
	Multiple cropping	Soybean—maize multiple cropping	(Xu et al., 2021; Souza et al., 2023)
Argentina	Intercropping	Sunflower||soybean, maize||soybean intercropping	(Andrade et al., 2012; Coll et al., 2012; Ojeda et al., 2018)
	Rotation	Wheat→soybean→maize, wheat→pea→maize→soybean rotation	(Salado-Navarro and Sinclair, 2009; Andrade et al., 2023)
	Multiple cropping	Wheat—soybean, barley—soybean, maize—pea, maize—soybean multiple cropping	(Andrade et al., 2017; Videla-Mensegue et al., 2021)

In farming system, the symbols '||', '→', and '—' represent intercropping, crop rotation across years, and multiple cropping within a year, respectively.

Table 2
Typical examples of improved agronomic efficiency of N (AE_N) and partial factor productivity of N (PFP_N) with crop diversification in countries with highly diversified cropping systems.

Country	Cropping system	Crop configuration	References
China	Intercropping	Wheat||faba bean, maize||potato, maize||pea, maize||soybean, sorghum-soybean intercropping	(Liu et al., 2021b; Chen et al., 2022; Xu et al., 2023; Zhao et al., 2024b; Wang et al., 2025a)
	Rotation	Wheat→maize, wheat→ rice, rice→oilseed rape, maize→ rice, rice→morel rotation	(Lu et al., 2021; Fu et al., 2023; Zhang et al., 2023; Wang et al., 2024c; Song et al., 2025)
	Multiple cropping	Winter wheat—summer maize, maize—rice, rice—green manure, spring wheat—green manure multiple cropping	(Liang et al., 2019; Han et al., 2022; Tong et al., 2022; Hu et al., 2024; Wei et al., 2025)
India	Intercropping	Maize||groundnut, maize||soybean, cotton||blackgram, pearl millet||pigeonpea intercropping	(Banik and Sharma, 2009; Munirathnam and Kumar, 2010; Jayakumar and Surendran, 2017; Kamdi et al., 2024)
	Rotation	Soybean→wheat, maize→wheat, rice-wheat, pigeon pea→vegetable mustard→okra rotation	(Aulakh et al., 2012; Sandhu et al., 2019; Bhardwaj et al., 2023; Garg et al., 2024)
	Multiple cropping	Rice—green manure, rice—wheat; wheat—green manure multiple cropping	(Pradhan and Mondal, 1999; Sharma et al., 2021; Kakraliya et al., 2024; Kumar and Bordoloi, 2024)
USA	Intercropping	Cabbage||eucalyptus torelliana, millet||soybean, garlic||oat intercropping	(Nissen et al., 1999; Jahanzad et al., 2015; Smychkovich and Hashemi, 2025)
	Rotation	Maize→oilseeds→soybean, maize→soybean, wheat→forage rotation	(Scott et al., 2021; Maas et al., 2022; Massigoge et al., 2024)
	Multiple cropping	Sweet maize—cover crop, spring wheat—pea multiple cropping	(Teasdale et al., 2008; Sainju, 2024)
Brazil	Intercropping	Maize||palisade grass, maize||congo grass, guinea grass||pigeon pea intercropping	(Almeida et al., 2018; Batista et al., 2019; Junior et al., 2020)
	Rotation	Pearl millet→maize, wheat→maize, maize→rice, maize→soybean rotation	(Rosolem et al., 2004; Crusciol et al., 2021; Galindo et al., 2021a; Otto et al., 2024)
	Multiple cropping	Green manure—common bean, cowpea—wheat multiple cropping	(Ferreira et al., 2013; Galindo et al., 2021b)
Argentina	Intercropping	Maize||soybean, peach tree||Lotus tenuis intercropping	(Chiesa et al., 2022; Ziegler et al., 2025)
	Rotation	Wheat→vetch plus oat, winter wheat→pea→vetch rotation	(Galantini et al., 2000; Salvagiotti et al., 2025)
	Multiple cropping	Wheat—soybean, barley—soybean, maize—pea, soybean—maize multiple cropping	(Andrade et al., 2017; Caviglia et al., 2019)

In farming system, the symbols '||', '→', and '—' represent intercropping, crop rotation across years, and multiple cropping within a year, respectively.

enhances WP, AE_N, and PFP_N in crop production systems. We consider the agronomically, physiological, and molecular aspects of such efficiency improvements focusing on interaction effects related to water and N dynamics. The review also examines the potential for fine-tuning crop diversification methods based on differences between day and night temperatures (DIFs) across various agroecological conditions and identify key knowledge gaps that require further research. Ultimately, this work seeks to provide a comprehensive framework for deploying crop diversification strategies in sustainable agriculture. In more precise terms, it addresses the efficient utilization of water and N resources through the integration of agronomy, plant physiology, soil science, and plant and microbial molecular biology principles.

2. Analysis of the current situation of improving WP, AEN, and PFPN through diversified planting

2.1. Diversified planting enhances WP of crops

Several planting methods like inter- or relay-cropping (Yin et al., 2015; Liu et al., 2025a), multiple-cropping (Dong et al., 2025), and crop rotation (Hussain et al., 2019), can notably increase the WP in crops. The methods optimize resource utilization by minimizing competition and maximizing soil moisture retention through successive planting (Masse et al., 2025) or co-cultivation of multiple crops (Yin et al., 2020b; Li et al., 2025a). Difference in growth habits and root systems allows plants to capture resources more efficiently with less waste and higher productivity (Lóczy et al., 2024; Wang et al., 2025b).

Research shows that besides improving WP, well-designed systems also increase resilience in rainfall-limited locations. For example, the cultivation of deep- and shallow-rooted plants together enables deep-rooted plants to access deep soil water, while the shallow-rooted plants access near surface soil water, thereby improving net water use efficiency (Yin et al., 2020b; Chai et al., 2021). In addition, the microclimate from the varied canopies lowers soil temperature and wind speed and reduces evapotranspiration rates (Ma et al., 2020; Dai et al., 2021). Consequently, farmers adopting diversified planting practices often achieve higher yields without increasing irrigation demand, making it a sustainable strategy for water conservation in agriculture.

2.2. Diversified planting enhances AE_N and PFP_N of crops

The improvement of AE_N and PFP_N through crop diversification has been well-established through numerous studies, mainly through optimization of N cycling and reduction of losses in agroecosystems (Dong et al., 2024b; Liu et al., 2024a). Unlike monocultures, which often lead to N depletion or leaching due to uniform crop uptake patterns, diversified systems utilize complementary plant traits to maximize N retention and availability (Xu et al., 2023; Demie et al., 2025). For instance, growing legume and cereal crops together can enhance AE_N and PFP_N with biological N fixation, whereby legumes transform atmospheric N into plant-accessible N, with the benefit accruing to companion crops (Dong et al., 2024b). Deep-rooted species in diversified rotations can also scavenge N leached below the root zone of shallow-rooted crops, thereby reducing groundwater pollution risks (Hadir et al., 2025; Lyu et al., 2024). The difference in crop growth rates and root exudates also

stimulates microbial activity and enhances N mineralization and diminishes reliance on N fertilizers (Jiang et al., 2022). Maize-soybean and maize-pea intercropping systems show increases in PFP_N of up to 34.5% compared with monoculture in semi-humid, arid-prone regions of China (Zhao et al., 2024b) and up to 45.2% in the oasis irrigation area of northwest China (Xu et al., 2023), demonstrating how strategic species combinations can yield synergistic nutrient-use advantages.

Diversified planting also aids in maintaining healthy soils in the long term, which in turn improves AE_N and PFP_N further, in addition to incremental N savings. Crop rotations with N-fixing species or cover crops reduce external fertilization applications through the conservation of soil organic matter and the promotion of N recycling (Dong et al., 2024d). For instance, rotation of maize and common vetch replenishes soil N naturally, while diversified root structures increase water-filled pore space, reduce compaction, and enhances aeration and N uptake efficiency (Lyu et al., 2024, 2025). Furthermore, diversified systems minimize N losses from volatilization and denitrification due to the retention of soil cover, which stabilizes soil moisture and temperature and optimizes microbial activity (Mooshammer et al., 2022; Lyu et al., 2024). In agroecological systems, integrating trees with annual crops (agroforestry) has also been shown to improve PFP_N, as tree litter provides slow-release organic N inputs (Rowe et al., 2005). Diversified planting shows promise in tackling the escalating concerns regarding fertilizer prices and environmental degradation, thus presenting a key strategy for environmentally sustainable N management with improved crop productivity and reduced ecological footprints. Hence, policy incentivization and farmers training in selecting optimal species combinations could accelerate adoption and transform diversified systems into a cornerstone in the implementation of climate-smart agriculture and the green transition in agriculture.

3. Mechanisms of improving WP, AEN, and PFPN through crop diversification

The practice of diversified planting to effectively enhance WP, AE_N, and PFP_N in crops has been proven by previous studies. The core mechanisms lie in reducing ineffective losses of water and N, improving soil physicochemical properties, optimizing interspecific interactions among crops, regulating plant physiological processes, and driving microbial communities toward beneficial water and N use (Fig. 1). In-depth analysis of these mechanisms provides important guidance for the precise design of diversified cropping systems.

3.1. Reducing ineffective losses of soil water and N

Crop diversification, in the form of intercropping, crop rotation, and multiple cropping, significantly reduces aboveground water and N losses and increases belowground resource capture. Notably, this happens due to the synergistic effects caused by canopy structure and root system behavior. The canopy in diversified fields produces a microclimate, which diminishes direct solar radiation, lowers soil surface temperature, and reduces evaporation (Rivest et al., 2022; Ma et al., 2023). Moreover, the existence of varied crop species changes the field's aerodynamic characteristics, decreasing the wind velocity at the ground and further decreasing evaporation (Chen et al., 2010b; Rivest et al., 2022). Interspecific competition in diversified fields also creates moderate water stress, which stimulates physiological changes in plants and maximizes the root-to-shoot ratio and limits luxury transpiration, which is a process in which plants loose excess without a corresponding increase in photosynthesis (Liu et al., 2025b; Sun et al., 2025). Diversified production systems enhance N retention through the modification of

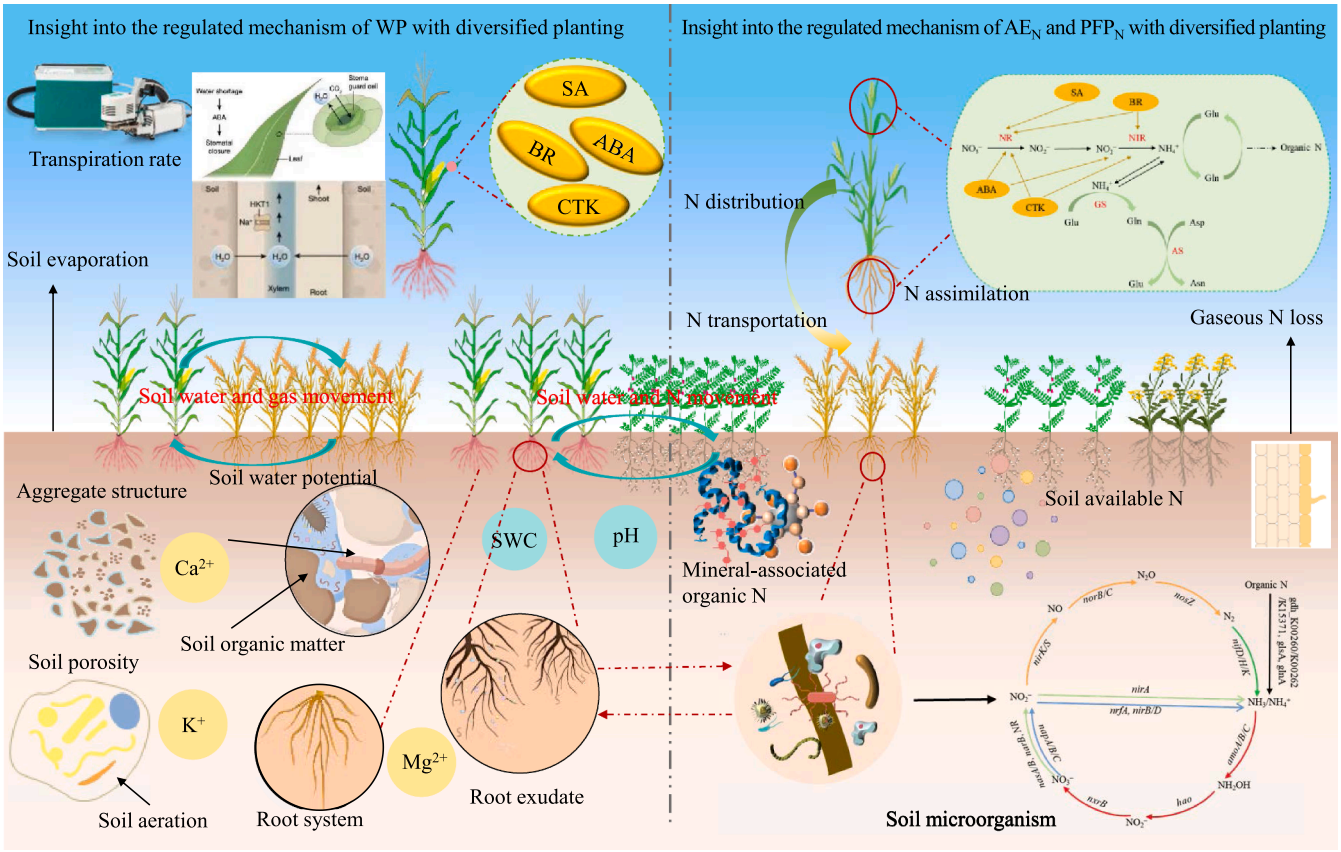


Fig. 1. Mechanisms of diversified cropping in improving water and nitrogen use efficiencies. AE_N, agronomic efficiency of N; PFP_N, partial factor productivity of N; WP, water productivity; SA, salicylic acid; ABA, abscisic acid; CTK, cytokinin; BR, brassinosteroid; SWC, soil water content; pH, soil pH; NR, nitrate reductase; NIR, nitrite reductase; GS, glutamine synthetase; AS, asparagine synthetase.

biogeochemical processing in the soil beyond conserving water (Nascimento et al., 2023; Gonzalez-Maldonado et al., 2025). The incorporation of legumes, for instance, maximizes biological N fixation and lessens the application of chemical-synthetic N fertilizers while conserving soil fertility (Dong et al., 2024c). Additionally, diversified systems influence soil pH and promote more heterogeneous microbial communities with significant effects on N cycling (Silva et al., 2020). Microbial activity in diversified crop fields enhances N immobilization in organic form and reduces losses through NH_3 volatilization and N_2O emissions (Wang et al., 2022b; Lyu et al., 2024). The superior structure of the soil in diversified farming maximizes infiltration and water retention and reduces leaching into deep-soil layers and groundwater (María et al., 2023; Yang et al., 2024). Generally, such mechanisms maximize the efficiency in the usage of water and N and minimize the losses through evaporation, leachates, and gaseous, while sustaining crop productivity.

One key advantage of having multiple crop species grow together is that it enables plants to absorb more water and N from the ground (Li et al., 2016; Singh et al., 2019). This reduces the need for external input. In contrast with monocultures in which all the roots in the same layers of soil compete for the same resources, diversified farming enables roots to exploit complementary soil niches. Crop roots in intercropping or rotation systems are revealed from research findings to possess wider spatial distributions, with some species having developed deeper roots and others extending laterally, enabling more complete soil exploration (Zhang et al., 2024b; Arafat et al., 2025). In such stratification, direct competition diminishes while total system productivity increases. Additionally, roots in diversified systems possess increased physiological activity, such as greater exudation of organic acids and enzymes that mobilize nutrients, resulting in greater availability of water and nitrate for uptake (Li et al., 2016; Bargaz et al., 2017). The presence of more than one species also triggers beneficial rhizosphere interactions, whereby mycorrhizal fungi and the N-fixing bacteria enhance nutrient acquisition (Liu et al., 2021a; Lyu et al., 2024). Aboveground diversified canopies maximize light interception with minimal mutual shading and longer photosynthetic periods (Sun et al., 2023; Luo et al., 2024). Enhanced photosynthetic efficiency translates into increased carbon allocation to roots, thereby improving their capacity for water and nitrate uptake.

Diversified systems also offer a sustainable biological foundation for high WP, AE_N, and PFP_N through the regulation of root structure and physiological processes such as expansion of the root uptake zone, extending uptake periods, and improve nutrient acquisition efficiency. Such processes are also useful for enhancing crop resilience under changing environmental conditions, as well as for promoting sustainable farming by reducing the reliance on chemical fertilizers and irrigation. In the long term, diversified crop production systems enhance soil health, mitigate greenhouse gas emissions, support climate regulation, and stabilize yields, thus forming a cornerstone of climate-smart and sustainable agriculture.

3.2. Improvement of soil physical and chemical properties

3.2.1. Optimizing soil aggregates structure and stability

Diversified planting systems greatly enhance WP, AE_N, and PFP_N by optimizing the structure and stability of soil aggregates, establishing a positive feedback loop between soil physical properties and plant resource capture (Zhang et al., 2022b; Akinola et al., 2023; Yang et al., 2024). Varied root structures and exudate profiles of multiple crop species in polycultures enhance the establishment of stable macro-aggregates (>250 μm) through the following multiple mechanisms: (1) fibrous root systems physically entangle soil particles (Shen et al., 2023), (2) taproots construct channels that facilitate the formation of stable aggregates (Bronick and Lal, 2005; Shen et al., 2023), and (3) varied carbon inputs from root exudates provide optimal substrates for aggregate-associated microbial communities (Jiang et al., 2022; Lin et al., 2023). Well-structured and stable aggregates improve water retention through enhanced pore connectivity and water-holding capacity, thereby increasing WP by suppressing unproductive evaporation and runoff (Jat et al., 2013; Soni et al., 2024). At the same time, protected microenvironments within aggregates conserve soil organic matter and promote microbial-mediated N cycling, thereby enhancing AE_N and PFP_N (Domnariu et al., 2024; Liu et al., 2024a). Importantly, the soil aggregate stability in diversified systems demonstrates excellent resistance to disruptive forces such as raindrop impact and tillage and sustains these advantages in the growing seasons (María et al., 2023; Soni et al., 2024). Hence, crop diversification can enhance the mean weight diameter (MWD) of the soil aggregates compared with monoculture and continuous cropping, thereby improving the soil aggregate stability and building the physical foundation for high WP, AE_N, and PFP_N.

The biological mechanisms for better aggregation in diversified systems produce synergistic outcomes on water and N dynamics through three main pathways. First, the various microbial consortia favored by diverse root exudates produce more glomalin-related soil proteins and extracellular polysaccharides-the 'glue' that binds aggregates (Roberson et al., 1995; Lu et al., 2025). Such biological bonding agents increase the proportion of water-stable aggregates and form pore networks that balance water retention and drainage (Jat et al., 2013; María et al., 2023). Second, the microhabitats in stable aggregates create spatially distinct aerobic and anaerobic zones that regulate nitrification and denitrification, minimizing N losses, while maintaining available N for plants (Tiemann et al., 2015; Bossolani et al., 2020). Third, the better structure facilitates root exploration, so plants can take up water and nutrients from a larger soil volume with less metabolic cost (Hadir et al., 2025; Arafat et al., 2025). The protected organic matter within aggregates from diversified systems also functions as a slow-release N reservoir, synchronizing N with crop demand (Locatelli et al., 2025; Lu et al., 2025). Such mechanisms account for why diversified systems experience higher productivity at the dual extremes of the rainfall-air temperature observed in production practices, with the resulting in superior yield stability compared with monocultures. Furthermore, diversified planting achieves sustainable yield improvements in WP, AE_N, and PFP_N beyond what can be obtained through chemical inputs alone, owing to biologically optimized soil architecture.

3.2.2. Enhancing the accumulation of soil organic matter and water retention capacity

Diversified planting systems significantly enhance WP, AE_N, and PFP_N through the progressive accumulation of soil organic matter (SOM), which serves as a cornerstone of key soil ecosystem functions (Domnariu et al., 2024; Liu et al., 2024a). The continuous input of diverse plant residues, including leaves, roots, and root exudates from different crop species, creates a heterogeneous organic matrix that supports robust microbial decomposition and humification processes (Mooshammer et al., 2022; Liu et al., 2024a). This diversity of organic inputs is particularly effective because it provides both rapidly decomposable compounds (e.g., legume residues) that fuel microbial activity and more recalcitrant materials (e.g., cereal straw) that build long-term SOM pools (Chai et al., 2021; Yang et al., 2024). As SOM content increases in diversified systems compared to monocultures, it directly improves water retention capacity through several mechanisms: each 1% increase in SOM enables the soil to hold an additional more than 15,000 additional liters of water per hectare, significantly boosting WP during drought periods (Hudson, 1994; Minasny and McBratney, 2017). Simultaneously, the N mineralization and immobilization turnover becomes more efficient, with SOM acting as both a reservoir for N storage and a slow-release mechanism that better matches crop demand patterns (Rowe et al., 2005; Locatelli et al., 2025). Field studies demonstrate that diversified cropping systems with enhanced SOM can reduce irrigation requirements while improving N recovery rates, creating a virtuous cycle of increasing WP, AE_N, and PFP_N over time (Drinkwater and

Snapp, 2007; Gardner and Drinkwater, 2009). Thus, diversified cropping systems can create self-reinforcing agroecosystems that continually enhance WP, AE_N, and PFP_N while reducing reliance on external inputs through intentional SOM accumulation driven by crop diversification.

The SOM's water retention advantages in diversified systems operate synergistically with processes of N cycling to provide optimal conditions for enhancing WP, AE_N, and PFP_N in crops (Drinkwater et al., 2017). SOM's sponge-like behavior augments plant available-water capacity while also modifying soil's hydraulic conductivity and maintaining optimal moisture conditions for microbial activity and root uptake (Or et al., 2007; Dexter et al., 2008). Moisture regulation becomes especially important for N transformations since it sustains the aerobic-anaerobic conditions required for effective nitrification and denitrification processes (Stark and Firestone, 1995; Groffman et al., 2009). In addition, SOM's colloidal structure increases cation exchange capacity, reducing N leachability while enhancing temporal alignment between N availability and crop growth stages (Groenigen et al., 2019; Kleber et al., 2021). SOM-associated soil fauna (e.g., earthworms and arthropods) further improve soil biological porosity, increasing infiltration rates two- to threefold compared with conventionally managed soils and reducing runoff and erosion (Jarvis et al., 2013; Groenigen et al., 2019). Evidence from the long-term agroecological research shows that diversified systems with perennials in combination with annuals with the potential for the most outstanding enhancements such that SOM accumulation rates from the range of 0.3 −0.8% annually can result in corresponding increases in WP, AE_N, and PFP_N (Syswerda et al., 2012; Lorenz and Lal, 2014). Such systems exhibit exemplary climatic resilience with the capacity for maintaining elevated productivity in low-rainfall years due to superior water retention and N delivery capabilities (Drinkwater et al., 2017; Lóczy et al., 2024). Thus, diversified cropping can create self-reinforcing agroecosystems that continually enhance WP, AE_N, and PFP_N while reducing reliance on external inputs through SOM accumulation driven by intentional crop diversification.

3.2.3. *Improving soil quality*

Intercropping and crop rotations substantially increase WP, AE_N, and PFP_N by improving soil health and soil quality through multiple interacting processes (Yan et al., 2023; Liu et al., 2025b). Combinations of multiple crop species maximize SOM accretion, microbial diversity, and soil structure, producing more fertile and stable growing environments (McDaniel et al., 2014b; Lange et al., 2015). Heterogeneous root profiles from deep taproots with shallow, branched roots, etc., increase the porosity and stability of soil aggregates and, through improved infiltrability and retention, more efficiently store and deliver water to roots (Jarvis et al., 2013; Lange et al., 2015). Modifying the soil structure in this way lessens evaporation from the soil and reduces runoff over the land surface, improving WP directly by making more water available for uptake (Jarvis et al., 2013; Rillig et al., 2014). Simultaneously, biochemical inputs from mixed crops broaden microbial functional diversity, enhancing N mineralization and reducing N losses from leaching and volatilization (Zak et al., 2003; Xu et al., 2024). In addition, suppression of soil pathogens and pests in diversified farming systems, reduces plant stress, allowing more plant resources to be allocated to growth rather than defense, thereby increasing WP, AE_N, and PFP_N (Latz et al., 2012; Pineda et al., 2017).

The long-term benefits of diversified planting in soil quality lead to sustained increases in WP, AE_N, and PFP_N, making such systems particularly useful in the face of climate variability (McDaniel et al., 2014b; Bowles et al., 2020). Ongoing input of fresh organic material from diverse crop residues and root exudates induces stable soil aggregates, which act to protect nutrients from premature leaching while at the same time maintaining favorable moisture regimes (Zhang et al., 2022b; Cibulková et al., 2025). In the longer term, this generates a 'soil memory' effect such that increased biological activity and nutrient cycling become self-reinforcing. For example, rotations involving legumes add N availability through biological N-fixation, thereby reducing dependency on synthetic fertilizers while maintaining high crop yields (McDaniel et al., 2014b; Wei et al., 2025). In addition, the buffering capacity inherent in high-grade soils reduces the effects of extreme weather; the effects of drought stress are reduced due to increased water retention, and excessive rainfall corresponds with less nutrient leaching (Thapa et al., 2018; Bowles et al., 2020). Empirical data obtained from the long-term experiments in the fields show that crop-rotation diversification can increase WP by up to 15% and PFP_N by up to 20% compared with monocropping over several decades in North America, with the difference growing more marked in adverse conditions (Bowles et al., 2020). In turn, agricultural producers can create productive and sustainable agroecosystems through crop diversification. These systems safeguard soil quality, thereby ensuring food security while also mitigating environmental degradation (Cui et al., 2018; Yang et al., 2024). In effect, this represents a paradigm shift from input-driven agriculture to ecologically intensified agriculture, where soil health underpins improvements in WP, AE_N, and PFP_N.

3.3. *Coordinated interspecific interactions*

3.3.1. *The spatio-temporal complementarity of the utilization of above-ground light resources*

Diversified planting systems notably raise WP, AE_N, and PFP_N through optimal spatial and temporal complementarity in the utilization of the above-ground light resource use (Brooker et al., 2014; Luo et al., 2024). The resulting complementarity derives from strategically designed crop combinations with differing canopy structure, growth habits, and phenological trajectories (Brooker et al., 2014; Chi et al., 2023). For example, growing tall, upright crops (e.g., maize) together with short, shade-compatible species (e.g., soybean or peanut) produces multi-layered canopies with high light capture and low competitive inhibition (Luo et al., 2024; Lu et al., 2025). The upper canopy layer formed by taller species captures direct solar radiation, while the understory crops efficiently utilize diffuse light, achieving elevated maximum photosynthetic productivity per hectare (Dong et al., 2024a; Lu et al., 2025). In the temporal dimension, the practices of strip intercropping, relay-planting, and multiple cropping enhance season-long light capture, while avoiding peak resource competition and light losses typical of monocultures (Ripoche et al., 2012; Wang et al., 2017). This maximized light allocation enhances photosynthetic assimilation and subsequently enhances WP with elevated biomass yield per amount of water transpired. In addition, enhanced photosynthetic activity stimulates root exudation and N uptake, thereby synergistically improving AE_N and PFP_N (Wang et al., 2017; Dong et al., 2024a). Hence, such spatially and temporally diversified planting arrangements can enhance WP, AE_N, and PFP_N over monoculture systems, and light resource partitioning remains essential in sustainable farming systems.

The capture of solar radiation in diversified planting configurations not only maximizes aboveground production but also triggers increased belowground growth (Hernández-Ochoa et al., 2022), leading to improvements in WP, AE_N, and PFP_N. Plants with efficient light capture produce more biomass, enabling the development of extensive, deep root systems that can explore a larger soil volume for water and nutrients (Postma and Lynch, 2012). In agroforestry configurations, deep roots of trees access water from deeper layers, while the shallower root systems of annual crops access moisture in the topsoil. Such mutual interaction among species reduces competition for available water and maximizes WP (Bayala and Prieto, 2019). Furthermore, enhanced carbon availability from photosynthesis stimulates soil microbial communities, thereby accelerating nitrogen release and improving AE_N and PFP_N (Sun et al., 2023; Huang et al., 2025). The differential light regimes inherent in varied canopy arrangements also help moderate local climatic conditions, leading to reduced soil evaporation and lower leaf temperature (Hernández-Ochoa et al., 2022; Demie et al., 2025). This conserves water and support plants physiological functioning. Importantly, the temporal dimension of light complementarity ensures that N

demand is distributed over time, preventing N losses that typically occur in monocultures with synchronized N demands (Ripoche et al., 2012; Hassen et al., 2017). Therefore, diversified cropping systems worldwide confirm that spatio-temporal light complementarity can lead to more resilient and resource-efficient agriculture, especially under climate variability. Harnessing these above- and below-ground synergies offers a powerful strategy for achieving sustainable intensification in global agriculture.

3.3.2. Ecological niche differentiation of root distribution and resource acquisition in the underground part

Ecological niche differentiation in root distribution and resource acquisition strategies can enhance WP, AE_N, and PFP_N in diversified planting systems (Lai et al., 2022; Song et al., 2022). Different crop species exhibit distinct root architectures (e.g., taproot *vs*. fibrous systems) and foraging behaviors that enable complementary exploitation of soil resources (Hauggaard-Nielsen and Jensen, 2005). For instance, deep-rooted species can access water and N from subsurface layers on 1 −2 m depth, while shallow-rooted crops efficiently capture resources in the topsoil on 0 −30 cm (Singh et al., 2019; Sun et al., 2022). This vertical stratification reduces direct competition and increases the cumulative volume of soil utilized, significantly improving WP, AE_N, and PFP_N at the system scale. Furthermore, root exudates from diverse species create chemically diverse rhizosphere environments that stimulate microbial communities involved in N cycling and organic matter decomposition (Jiang et al., 2022; Liu et al., 2024a). Using isotopic labeling methods, studies have shown that in diversified systems based on leguminous crops, N interception rates increase by 20 −40% compared with monocultures in Europe, while water uptake is improved due to minimal overlap in root depletion zones (Pirhofer-Walzl et al., 2011; Brooker et al., 2014). Niche complementarity also operates over time, as long- and short-season crops develop root systems at different stages, sustaining continuous resource uptake during the growing season (Postma and Lynch, 2012; Wang et al., 2025b). Such specialization is especially beneficial in water-limited and nutrient-poor conditions, significantly enhancing WP, AE_N, and PFP_N compared with monocultures.

The benefits of root niche differentiation are also amplified through facilitative interactions with soil biota, generating a self-perpetuating feedback process that enhances WP, AE_N, and PFP_N (Bender et al., 2016; Steinauer et al., 2016). Mycorrhizal networks, for example, integrate root systems of co-planted species and form shared hyphal pathways that transport water and N in accordance with plant demand (Egerton-Warburton et al., 2007). In cereal−legume mixtures, the networks facilitate the uptake of N from N-fixing leguminous crops and transport it to cereal crops while also augmenting water transport under drought conditions (Adams et al., 2017; Duchene et al., 2017). In addition, the diverse root exudate profiles in multiple crops selectively promote beneficial microbial communities, which including N-fixing bacteria and phosphate-solubilizing fungi increase access to nutrients (Steinauer et al., 2016; Zhou et al., 2023). Physically, root-driven soil reconfiguration enhances both hydraulic and structural properties, with taproots creating biopores that increase infiltration rates while fibrous roots promote aggregation and reduce evaporation (Jarvis et al., 2013; Rillig et al., 2014). A field experiment demonstrates that wheat–maize intercropping with straw retention increases leaf water potential and photosynthetic rates under stress, due to such optimized belowground interactions (Yin et al., 2021). Long-term studies confirm that root niche differentiation reduces N demand and improves drought tolerance supporting sustainable intensification under climate change (Zuppinger-Dingley et al., 2014; Brooker et al., 2014). Therefore, diversified crop systems achieve unprecedented efficiencies in N and water utilization that are inconceivable in monocultures with the aid of spatial partitioning in root zones and also their dynamic interactions with soil biology.

3.3.3. The interaction between above- and below-ground processes in diversified planting systems

Diversified planting practices maximize WP, AE_N, and PFP_N through dynamic feedback between above- and below- ground processes (Nissen et al., 1999; Zhang et al., 2022b). The three-dimensional canopy structure aboveground optimizes light interception and regulates the microclimate, while the root secretions and microbial activities in the underground promote nutrient cycling and water retention (Hernández-Ochoa et al., 2022). For example, intercropping tall cereals with short legumes creates a multi-layer canopy. This structure maximizes light energy utilization while reducing soil evaporation. Simultaneously, the deep-rooted cereals and N-fixing legumes facilitate subsoil water acquisition and N-fixation in the rhizosphere, respectively (Li et al., 2009; Luo et al., 2024). These interactions are mediated by plant-plant signaling and resource partitioning, where species with complementary traits (e.g., C_3 and C_4 photosynthesis, taproots compared with fibrous root systems) reduce competition and increase system productivity (Brooker et al., 2014; Zhou et al., 2023). Thus, above-belowground synergies increase WP, AE_N, and PFP_N through integrated partitioning of light, water, and nutrient niches.

The interaction among above- and below-ground processes in varied cropping systems generates self-enhancing feedbacks that sustain high WP, AE_N, and PFP_N (Nissen et al., 1999; Zhang et al., 2022b). Aboveground biomass production supports carbon allocation to roots, which in turn stimulates soil organic matter and microbial community, leading to increased water retention and N provision (Lange et al., 2015). For instance, legumes used in some cropping systems capture atmospheric N and then transfer N later to the adjoining cereals in root exudates or mycorrhizal links, thus decreasing the use of chemical N fertilizers (Li et al., 2009; Huang et al., 2025). Meanwhile, shade from the tall plants decreases the temperature and evaporation at the soil level and aids shallow-rooted species while stimulating deeper root growth for access in drier stretches (Singh et al., 2019; Sun et al., 2022). Feedbacks are reinforced with the interaction among plants and microbes, where microbes in the rhizosphere dissolve nutrients and augment tolerance to distress (Berendsen et al., 2012; Parks and Friesen, 2025). Thus, diversified farming systems with strong above- and below-ground interactions are better able to withstand climatic extremes and maintain stable yields despite fluctuations in water and nitrogen availability. This resilience positions them as a cornerstone of sustainable agriculture. This approach manages system interactions to balance crop productivity with resource conservation, primarily through diversified planting.

3.4. The physiological regulatory mechanism

3.4.1. Optimization of source-sink relationship and adjustment of carbon allocation

The optimization of the source-sink relationships in diversified crop systems optimizes WP, AE_N, and PFP_N via balanced allocation of photosynthetic assimilates in organs for growth and storage (Zhang et al., 2020; Yang et al., 2025a). In monocultures, imbalanced carbon and nitrogen partitioning often leads to inefficient resource use, whereas diversified systems enable more efficient coordination of carbon-nitrogen allocation (Lai et al., 2022; Lu et al., 2025). For example, legume crops in diversified agroecosystems are capable of fixing N from the atmosphere and therefore improving the availability of N, while adjacent graminaceous crops are good carbon sinks and exploit such N more competitively in terms of biomass yields (Zhang and Li, 2003; Hauggaard-Nielsen et al., 2008). This interaction reduces N loss and increases photosynthetic efficiency. Additionally, diversified cropping systems enhance hydraulic redistribution, where deep-rooted crops draw water from lower soil layers, benefiting shallow-rooted companion-crops through hydraulic lift (Yin et al., 2018; Singh et al., 2020). This process optimizes water use while maintaining high carbon assimilation rates. Accordingly, diversified planting schedules have increased stomatal conductance and net photosynthesis rates with water

limitation due to improved N availability and hydraulic control, thus improving WP, AE_N, and PFP_N.

A key biological process driving improvements in WP, AE_N, and PFP_N in diversified planting is the adjustment of plant carbon and nitrogen allocation in response to interspecific interactions (Brooker et al., 2014; Zhang et al., 2019). In diversified farming system, crops adjust growth strategies and allocate carbon that would otherwise support excessive leaf growth in monocultures toward reproductive structures and storage organs (Lange et al., 2015; Dong et al., 2024a). Doing so enhances the volume of crops used and retains nutrients in the soil better in diversified planting arrangements. For instance, planting legumes with cereals reduces competition for N so that the cereals can allocate more carbon in producing grains instead of producing more roots (Zhang et al., 2019). Furthermore, the roots from different types of plants exude chemicals that promote microbial activity, and the result breaks up the organic material and releases the N in ways that facilitate superior carbon-nitrogen links improving the WP, AE_N, and PFP_N (Duchene et al., 2017; Zhang et al., 2019). High efficiency in transferring N in the diversified arrangements ensures the N ends up in the growing seed or fruit instead of being lost with age or washed away. Such biological modifications, in addition to favorable growing conditions offered from the plant canopies, result in sustained high WP, AE_N, and PFP_N in diversified farming arrangements. Therefore, diversified planting gives a way to sustainably increase crop yield in settings where resources are limited by optimizing the balance of sources and sinks and the allocation of carbon and N.

3.4.2. *Interspecific differences in stomatal behavior and water use strategies*

Diverse planting systems maximize WP through the exploitation of interspecific differences in stomatal regulation and water use strategies among associated species (Brooker et al., 2014; Li et al., 2025b). Various species possess unique stomatal behaviors. Some species employ conservative water-saving tactics with low stomatal conductance and deep root systems, whereas others focus more on carbon capture with high transpiration rates and shallow root systems (McDowell et al., 2008; Pelech et al., 2021). When appropriately combined, such complementary characteristics maximize total water use tactics. For instance, in legume||cereal intercropping, legumes generally close their stomata partly and cut back water loss, thus maintaining stable water use at the whole system scale, while the cereal crops balance photosynthesis with high stomatal conductance and moderate water usage at moderate water deficits (Sekiya and Yano, 2004; Li et al., 2019). Hydraulic redistribution assisted by deep-rooted species also enhances soil moisture supply to shallow-rooted neighbor species, enhancing WP (Yin et al., 2018; Singh et al., 2020). Such adaptive stomatal attributes induced through species interaction allow plants to take maximum advantage of the fraction of carbon assimilated per amount of used water, especially in changing environmental conditions.

The stomata behave behavior and how plants efficiently use N in diversified planting systems come from different strategies of species for getting water and N (Lai et al., 2022; Lu et al., 2025). Such tactics facilitate a more equitable allocation of resources. For instance, the coincident evolution of C_3 and C_4 plants evokes contrasting reactions in the face of variations in light and humidity. Such differential behaviors enable such plants to tap the available resources more efficiently at the temporal and spatial scales (Yin et al., 2018; Zhang et al., 2022b). In the case of a maize||soybean intercropping crop system, the C_4 maize plants growing together with soybean typically exhibit elevated WP due to their high CO_2 assimilative efficiency. On the other hand, the C_3 soybean plants growing with maize benefit from the shade from the taller-growing maize plants and experience reduced stress from the water, in addition to reduced demand for N (Li et al., 2019; Sun et al., 2023). Such functional complementarity minimizes competition and enhances AE_N and PFP_N by channeling N toward growth rather than stress responses. Consequently, these interactions further improve AE_N and PFP_N. In addition, the incorporation of N-fixing legumes in diversified cropping systems elevates soil N levels, which in turn alters stomatal behavior. This enhanced N status increases stomatal sensitivity to drought conditions, thereby improving water use efficiency (Zhang and Li, 2003; Hauggaard-Nielsen et al., 2008; Brooker et al., 2014). Thus, diversified cropping systems form a robust physiological platform for enhancing crop production in the arid areas with the enhancement of WP, AE_N, and PFP_N through the tapping of in-built differences in stomatal behavior and resource usage.

3.4.3. *The transduction and resilience of adversity signals have been enhanced*

Diversified planting systems improve crop resilience and resource use efficiency by enhancing perception and response to environmental stressors, particularly those related to water and N (Brooker et al., 2014). Unlike monocultures, plants in mixed systems can detect and respond to stress cues more rapidly and effectively (Xiong et al., 2013). For example, root exudates from companion species may signal neighboring plants to adjust stomatal behavior or water consumption before severe soil desiccation occurs (Hernández-Ochoa et al., 2022; Zhou et al., 2023). Such inter-plant communication through chemical and hydraulic signaling improves collective water management and WP. Similarly, N stress signals are managed more effectively in diversified planting systems due to complementary N acquisition strategies among species. For example, legumes have the capability to share N-rich compounds with non-leguminous plants through mycorrhizal networks, which aids in mitigating N deficiency and curtailing unnecessary excess consumption (Pirhofer-Walzl et al., 2011; Brooker et al., 2014). Consequently, crops in diversified systems sustain higher photosynthetic rates during drought by sensing stress signals and optimizing the trade-offs between carbon uptake and the use of water and N, thereby minimizing productivity losses.

The ability of crops to tolerate limitations due to water and N in intercropping and rotation systems can be due to increased stress tolerance mechanisms and increased interspecies communication (Brooker et al., 2014; Zhou et al., 2023). In monoculture systems, a stable stress tolerance can weaken the whole system's robustness. However, in the case of diversified crop systems, they have fluctuating stress tolerances across species, which can assist in maintaining productivity in some parts of the system, even in unfavorable conditions (Brooker et al., 2014; Zuppinger-Dingley et al., 2014). For instance, an early-flowering crop species can develop ways to escape from drought, while some species with deep root systems can use tolerance mechanisms in order to tolerate drought, together helping in maintaining ecosystem productivity (Sekiya and Yano, 2004; Ouattara et al., 2023). On a molecular level, crops in diversified planting systems have increased expression of stress-responsive genes like aquaporins and nitrate transporters because of help from other species, which leads to better use of water and N (Ouattara et al., 2023). Also, different canopy types create a physical environment that reduces soil evaporation and moderates soil temperature fluctuations, thereby helping to ameliorate plant stress signals (Yin et al., 2020b). Therefore, diversified cropping systems act as a buffer against climate variability, strengthening physiological resilience, stabilizing yield, and promoting the long-term sustainability of climate-smart agriculture by improving the reliability and resilience of stress signal processes.

3.5. *The mechanisms driving microbial activity*

3.5.1. *Rhizosphere microbial community structure and functional shifts*

Diversified planting systems, including multiple cropping, rotation, and intercropping, substantially restructure rhizosphere microbial communities, enhance waters and N use efficiency, and thereby, increase WP, AE_N, and PFP_N (Wahbi et al., 2016; Jiang et al., 2022; Masse et al., 2025). These systems enhance soil microbial activity and diversity through the input of diverse root exudates and crop residues. They enhance beneficial microbial communities like *Proteobacteria* and

Bacteroidetes, which are highly useful in nutrient recycling and stress coping (Vukicevich et al., 2016). For example, legume crops increase the abundance of N-fixing microbes and mycorrhizal fungi in a diversified crop system. It helps in availing more N and allows for better uptake of waters with their synergies (Qiao et al., 2024). In addition, the diversified planting systems reduce the extent of pathogenic microorganisms while building up a microbial community allowing resource sharing, such as the uptake of deep-rooted and shallow-rooted plants, allowing easier access to water and nutrients (Singh et al., 2019). Genetic analyses of these systems have identified the enrichment of key functional genes, including those responsible for cellulose degradation (cellulases) and N-fixation (*nifH*). Concurrently, a reduction in nitrification-related genes (*amoA*) promotes nitrogen retention in the soil, thereby improving nutrient use efficiency.

Synergistic effects on WP, AE_N, and PFP_N in diversified systems are regulated through microbially driven carbon and nitrogen cycling processes (Hao et al., 2022; Yan et al., 2023). Intercropping legumes with cereals, for instance, increases the *nifH* gene abundance in the rhizosphere, thereby increasing N inputs directly while decreasing reliance on the application of chemical fertilizers (Li et al., 2009; Huang et al., 2025). At the same time, diversified planting systems enhance SOC content through the augmentation of root biomass and root exudates, facilitating microbial carbon use efficiency and the stabilization of soil moisture, making it reliable in enhancing WP (Yan et al., 2023; Hadir et al., 2025). Hence, crop diversification enhances climatic adaptability by leveraging microbial mechanisms that improve drought tolerance and nitrogen retention, thereby resolving the trade-off between sustainable yield and efficient water and N use.

3.5.2. *The promoting effect of mycorrhizal fungal networks on water and N transport*

Diversified planting systems help to increase WP, AE_N, and PFP_N by fostering symbiotic relationships with arbuscular mycorrhizal fungi (AMF), which form extensive hyphal networks that facilitate the transfer of water and nitrogen (Hodge and Fitter, 2010; Zhang et al., 2022a). AMF attach to plant roots and spread their threads well beyond the roots, which helps plants reach more soil for water and N (Hodge and Fitter, 2010). These fungal networks work like natural fertilizers, improving how plants take in N by changing organic N into forms that plants can use and lowering the loss of N in water (Hodge and Fitter, 2010; Cavagnaro et al., 2015). Furthermore, AMF contribute to improving WP by creating a good soil structure, holding more water, and allowing the plants to absorb water from deeper soil layers at dry periods (Wang et al., 2024b). Growing multiple types of plants in diversified cropping systems produces a more complete and vibrant community of AMF compared with growing a single crop, since multiple crops provide food for different fungi (Guzman et al., 2021). In specific terms, legumes in multiple crops assist in stimulating nitrifying and N-fixing bacteria, which cooperate with the AMF in producing the cycling of N well and improving the use of N (Liu et al., 2021). As a result, crop associated with AMF in diversified cropping systems exhibit improved WP, AE_N, and PFP_N, highlighting the critical role of microbial interactions in sustainable crop production.

The interface among varied systems of cropping and mycorrhizal fungi provides a beneficial cycle enhancing WP, AE_N, and PFP_N (Wahbi et al., 2016; Guzman et al., 2021). AMF networks function as subterranean nutrient transfer pathways, redistributing water and N among plants based on demand, referred to as the 'common mycorrhizal network (CMN)' effect (Ding et al., 2022). In intercropping or crop rotation systems, N-fixing legumes can transfer surplus N to neighboring or following plants through shared mycorrhizal fungal connections (Li et al., 2009; Xia et al., 2023; Huang et al., 2025). This transfer reduces the N demand of the receiver plants. Moreover, AMF assist in enhancing soil structure in varied systems of crops, which decreases N loss and enables the penetration of water into the soil, additionally enhancing increased WP, AE_N, and PFP_N (Cavagnaro et al., 2015; Delavaux et al., 2017). Diversified planting systems also promote varied types of bacteria assisting in the decomposition of N, enhancing AE_N and PFP_N through facilitating nutrient transport with the assistance of AMF (Yan et al., 2023; Zhu et al., 2023). Thus, crops grown in AMF-rich diversified systems produce higher yields and have better WP, AE_N, and PFP_N in dry conditions compared to single-crop systems, demonstrating the benefits derived from these microbial partnerships.

3.5.3. *Microbial processes of biological N fixation and N transformation*

Diversified planting systems incorporating leguminous crops significantly enhance biological nitrogen fixation through synergistic interactions between plants and diazotrophic microorganisms (Peoples et al., 2009; Zhao et al., 2022). Incorporation of legume crops in rotation or in intercropping systems allows ecological niches for root-nodule N-fixing bacteria like *Rhizobium*, *Bradyrhizobium*, and *Azospirillum* that possess the enzyme nitrogenase (*nifH* genes) in the process of converting ambient N_2 into plant-accessible ammonium N (Oldroyd, 2013; Latati et al., 2016). Metagenomic analysis indicates that diversified systems can significantly increase diazotrophic activity compared with monoculture of single crop species, due to root exudates (e.g., flavonoids, organic acids), which selectively support the enrichment of diazotrophs (Venter et al., 2016; Zhalnina et al., 2018). For instance, soybean/fababean||maize intercropping elevates *nifH* gene expression in rhizosphere soils, contributing 25–50 kg N ha^{-1} annually via biological N fixation (Li et al., 2016; Lin et al., 2024). In addition, non-legume crops like cereals benefit from associative N fixation by free-living diazotrophs (e.g., Azotobacter), particularly when intercropped with phosphorus-mobilizing crops that fuel microbial activity with carbon-rich root exudates (Kennedy et al., 2004; Bhattacharjee et al., 2008). Thus, the interactions in diversified systems diminish the reliance on synthetic chemical N fertilizers while enhancing AE_N and PFP_N through site-specific N delivery at the root zone.

A structurally diversified planting pattern can optimize processes like N mineralization, nitrification, and denitrification through shifts in microbial community composition (McDaniel et al., 2014b; Dong et al., 2024b). Functional guilds like Nitrosomonas (ammonia oxidizers) and Nitrobacter (nitrite oxidizers) are suppressed in diversified planting systems, which leads to reductions in nitrification rates reduction and subsequent N leaching by 20–35% in grain crop production in temperate agroecosystems (Gardner and Drinkwater, 2009). Credit goes to: (1) allelopathic metabolites (e.g., sorgoleone from sorghum) that suppressed nitrifier activity (Subbarao et al., 2015), and (2) elevated carbon availability from multiple root exudates, which enhances heterotrophic microbial immobilization of NH_4^+ (Duchene et al., 2017; Jiang et al., 2022). At the same time, these systems enhance N mineralization by promoting Actinobacteria and Bacillus populations, which rapidly break down organic N into plant-consumable N forms (María et al., 2023; Holatko et al., 2024). Like legume→cereal rotation or legume||cereal intercropping, protease and urease activities are increased and thus soil N release and crop N demand are coordinated (Restovich et al., 2012; Xu et al., 2023). Such N cycling precision enhances AE_N and PFP_N while reducing N_2O emissions through modulation of denitrifier activity and lower *nosZ* gene abundance (Xu et al., 2023; Lyu et al., 2024). Thus, diversified systems create dual benefits for water and N, mycorrhizal networks such as Glomus spp. enhances water absorption of the roots, while stabilized soil N pools minimize the losses caused by irrigation and heavy rainstorms. Together, these mechanisms highlight diversified planting system as a key strategy for sustainable intensification for increasing WP, AE_N, and PFP_N in resource-limited conditions.

4. Optimal design and regulation approaches of diversified planting systems for improving water and nitrogen use efficiencies of crops

Crop diversification is a vital pathway to promoting sustainable agricultural development and green transformation in arid and semi-

arid regions as well as in areas with limited arable land per capita. The WP, AE_N, and PFP_N of crops in diversified planting systems can be effectively enhanced by optimizing crop combinations and configurations, innovating water and fertilizer management techniques, improving tillage and mulching practices, and integrating smart monitoring and decision support systems (Fig. 2). These mechanisms, which link agronomic practices to improved water and nitrogen outcomes, are conceptually illustrated in Fig. 3, providing a holistic framework for understanding how diversified systems enhance water productivity and nitrogen use efficiency. A comprehensive analysis of various scientific findings and an in-depth exploration of optimized design and regulation pathways for crop diversified systems to improve WP, AE_N, and PFP_N are of great significance for advancing both research and application in field practices. From practice to outcomes schematic diagram illustrating the mechanisms of diversified cropping systems in enhancing water productivity and nitrogen use efficiency (Fig. 3).

4.1. Crop combination and configuration optimization

4.1.1. The principle of crop combination based on complementary functional traits

Growing varying crop types together, which possess characteristics that benefit each other, can raise the efficiency of water and nutrient use in agriculture (Duchene et al., 2017; Zhang et al., 2022b). The point is to select crop types or crop varieties which exploit resources in varying but beneficial ways, such as having alternative root structures, phenological characteristics, and ways of absorbing water and nutrients (Sekiya and Yano, 2004; Bargaz et al., 2017). For instance, deep-rooted crops such as maize or sorghum can access shallower-water and N from deeper in the soil profile, while shallow-rooted legumes such as peas or soybeans capture N from the atmosphere, thus less chemical N fertilizer (Postma and Lynch, 2012; Tan et al., 2021; Shen et al., 2023). Intercropping early- and late-season crops enables the more efficient gathering of resources through the lessening of competition while in maturation stages (Hauggaard-Nielsen et al., 2008; Yin et al., 2018; Hu et al., 2020b). Organization of the crops, such as in the spacing between strips or in the pattern of the row in an intercropping pattern, can lead to superior growing conditions, less evaporation from the soil, and the preservation of N in the soil (Hu et al., 2020a; Tan et al., 2021). Smallholder producers can thus use water and nutrients more proficiently and increase productivity while preserving the environment.

4.1.2. Spatial layout and time series design

The crop layout in some planting methods, including plant- and row-spacing, actually does count in improving WP, AE_N, and PFP_N (Fan et al., 2020; Zhang et al., 2020; Tan et al., 2021). For example, if the maize and

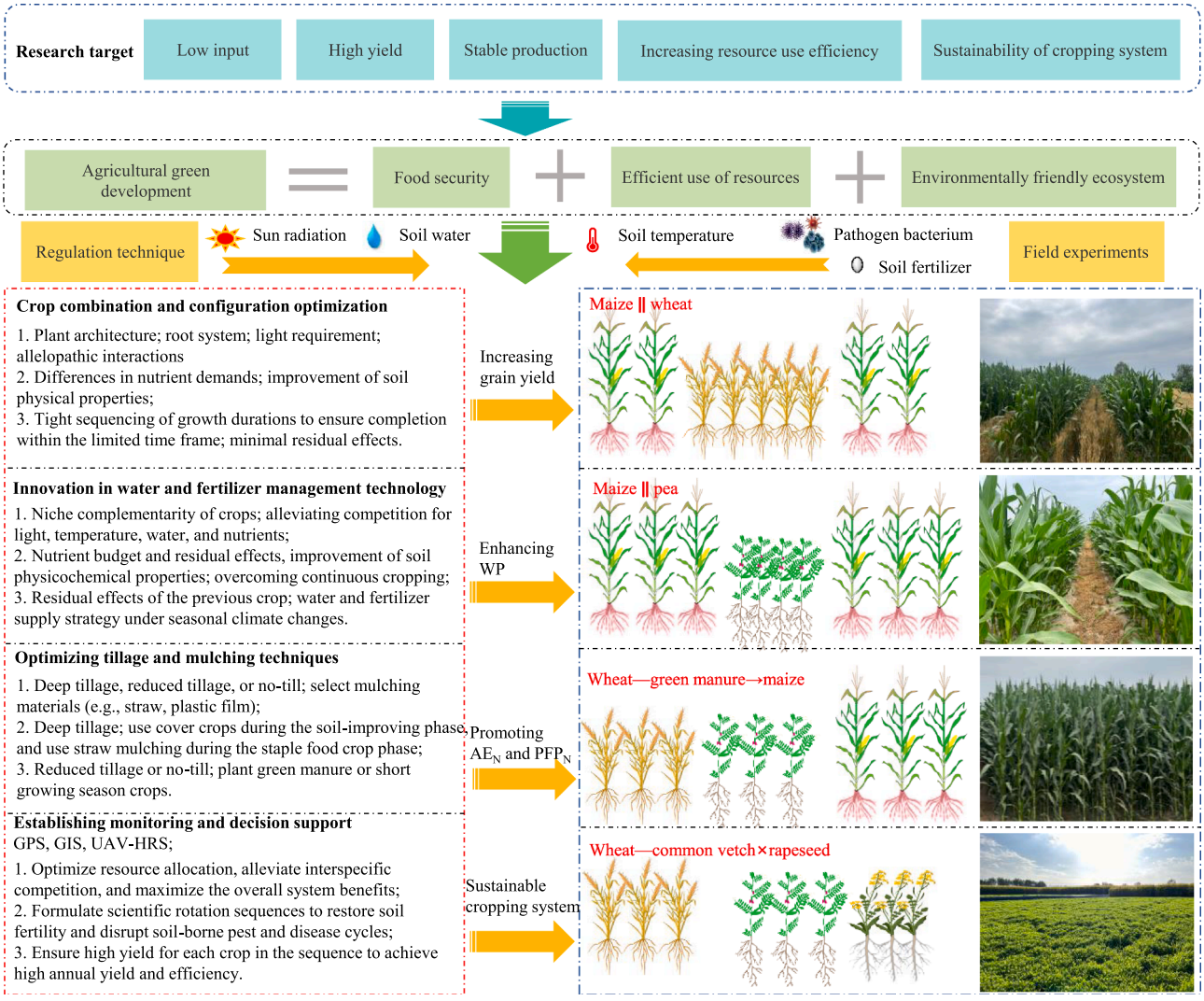


Fig. 2. Design and regulation mechanism for sustainable diversified cropping: Achieving high yield and efficient utilization of water and N. GPS, global positioning system; GIS, geographic information system; UAV-HRS, unmanned aerial vehicle hyperspectral remote sensing; AE_N, agronomic efficiency of N; PFP_N, partial factor productivity of N; WP, water productivity; ‖, intercropping; →, crop rotation; —, multiple cropping; × , mixed sowing.

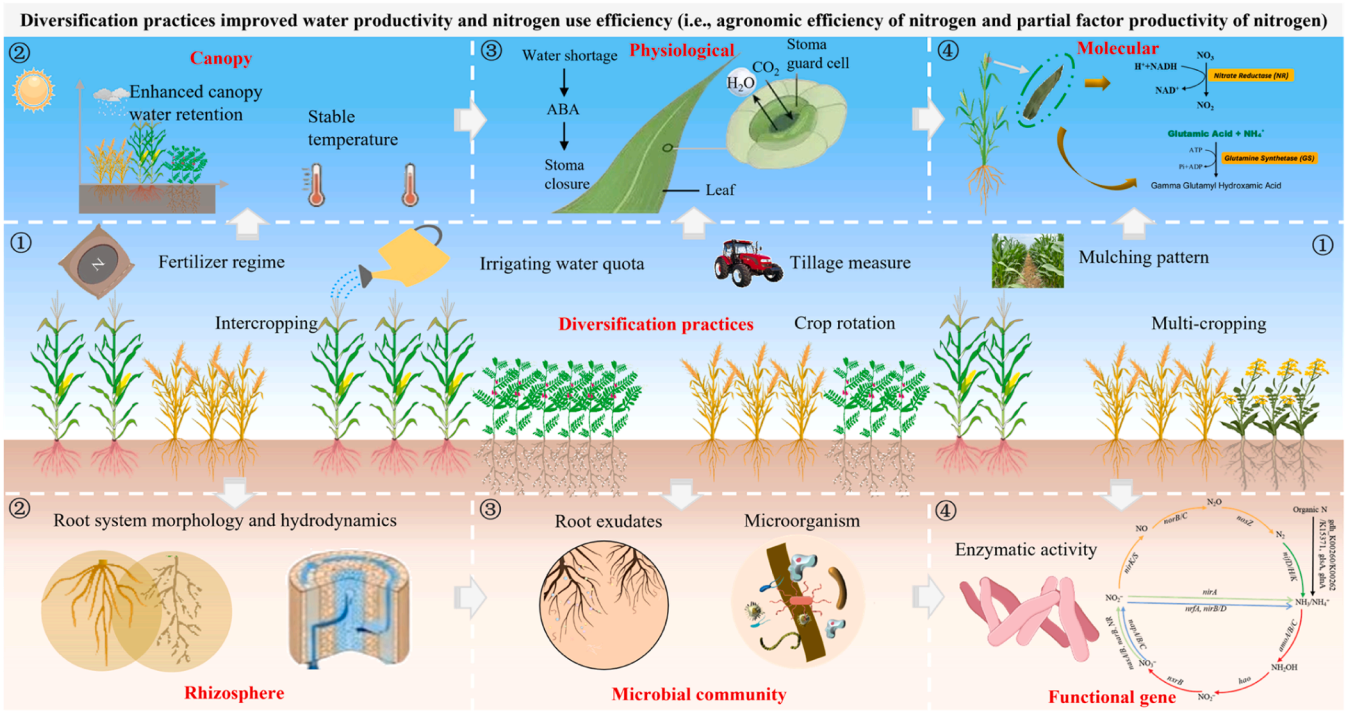


Fig. 3. From practice to outcomes framework illustrating the mechanisms of diversified cropping systems in enhancing water productivity and nitrogen use efficiency. This conceptual model delineates the cascade of effects initiated by (1) crop diversification practices (intercropping, crop rotation, and multi-cropping) coupled with targeted field management (fertilizer regimes, irrigating water quotas, tillage measures, and mulching patterns). These practices drive (2) spatial and temporal biophysical alterations, including enhanced canopy water retention and temperature stability aboveground, alongside optimized root system morphology and hydrodynamics within the rhizosphere. Subsequently, these changes trigger (3) interconnected physiological and microbially-mediated regulatory processes. Physiologically, plants adapt to constraints such as water shortages through ABA-mediated stomatal closure. Concurrently, altered root exudates dynamically shape the rhizosphere microbial community. At the (4) molecular and functional gene levels, these adaptations are governed by key nitrogen assimilation enzymes—specifically nitrate reductase (NR) and glutamine synthetase (GS), as well as soil microbial enzymatic activity and nitrogen-cycling functional genes (e.g., *nifH/K*, *amoA/B/C*, *nirK/S*, and *nosZ*). Ultimately, these multi-tier mechanistic pathways converge to generate positive impacts, significantly improving overall water productivity and nitrogen use efficiency (i.e., agronomic efficiency of nitrogen (AE_N) and partial factor productivity of nitrogen (PFP_N)). Abbreviations: WP, Water productivity; NUE, Nitrogen use efficiency; ABA, Abscisic acid; NR, Nitrate reductase; GS, Glutamine synthetase; AE_N, Agronomic efficiency of nitrogen; PFP_N, Partial factor productivity of nitrogen.

soybean are close together in strip intercropping system, more light is harvested and less evaporation from the soil, thereby improving WP (Rahman et al., 2016). On the other hand, planting deep-rooted and shallow-rooted crops in certain ways allows more effective utilization of water and N in the upper and subsoil layers, also improving WP, AE_N, and PFP_N (Singh et al., 2020; Tan et al., 2021). For example, alternating the tall cereal crop maize with the low-growing leguminous crop peanut in strip intercropping can reduce competition while maximize the fixation of N and absorption of water (Jiang et al., 2022; Zhao et al., 2022), and thus facilitate the efficient use of water and N. Furthermore, changing the planting density in the row can also optimize resource use. Higher planting density in arid areas enhances WP by promoting canopy shading and reducing soil evaporation, whereas in N-rich soils, lower density minimizes competition for other resources (Fan et al., 2020). Precision agriculture technologies such as GIS-based spatial analysis can optimize layout by accounting for field heterogeneity (Zhou et al., 2009), so that planting in crop fields can be in accordance with the soil moisture holding capacity and the available amount of N. This clearly promotes the synchronization in crop demand with the supply of soil water and N and thus facilitates the efficient use of them.

The temporal aspect of diversified planting systems, specifically crop rotation cycles, is no less crucial in maintaining high WP, AE_N, and PFP_N in the long run (McDaniel et al., 2014a; Massigoge et al., 2024; Zhao et al., 2024a). Well-designed rotations break pest and disease cycles, improve soil structure, and increase organic matter content, thereby enhancing N mineralization and water retention (Tiemann et al., 2015; Bowles et al., 2020). For illustration, alternating high-N-demanding cereals with N-fixing legumes enables a more balanced N budget and lessens N dependency (Jayakumar and Surendran, 2017; Wei et al., 2025). Short-duration cover crops reduce N leaching, enhance off-season soil moisture retention, and improve both WP, AE_N, and PFP_N (Wang et al., 2024a; Wei et al., 2025). Furthermore, crop sequence should also consider their differing root exudates and characteristics in residue decomposition, which are beneficial in enhancing WP, AE_N, and PFP_N through the promotion of beneficial soil microbe activity and the maximization of the water and N cycle (Hu et al., 2018; Tong et al., 2022; Dong et al., 2025). Dynamic rotation planning with the aid of modeling tools that project the dynamics among soil-water-nitrogen along the long term can set up the best rotation sequence for high WP, AE_N, and PFP_N in varied agroecological zones (Syswerda et al., 2012). For example, in rainfall-limited environments, alternating drought-resistant crops (e.g., millet) with deep-rooted perennials (e.g., alfalfa) can thereby stabilize yield and elevate WP, AE_N, and PFP_N with minimal irrigation (Wang et al., 2006; Jia et al., 2009). Hence, farmers can adopt more resilient and resource-efficient cropping systems by synchronizing crop rotation with seasonal rainfall, irrigation regimes, and nutrient release dynamics (Bowles et al., 2020; Massigoge et al., 2024). However, the success of crop combination and spatial–temporal layout is highly dependent on environmental factors, and inappropriate configurations can lead to severe resource use inefficiencies. Under water resource constraints or low soil fertility, intense interspecific competition for limited water and N may outweigh the facilitative benefits of an intercropping system, resulting in yield reductions for both crops (Nissen et al., 1999; Singh et al., 2020). Furthermore,

complex spatial configurations (e.g., strip intercropping) often face practical limitations, including increased labor costs and a high incompatibility with large-scale mechanized planting and harvesting (Cui et al., 2018). Therefore, the implementation of diversified cropping must be guided by local climatic conditions, water availability, soil fertility, and mechanization capacity, rather than being treated as a universally applicable solution.

4.2. Innovation in water and fertilizer management technology

At the crop production level, the imbalance in water and N inputs strongly affects crop yields and environmental safety (Sinclair and Rufty, 2012). Over-irrigation increases N leaching and accelerates groundwater contamination, while excessive application of N triggers luxury transpiration and waste of water and nutrients, and subsequently reduces WP, AE_N, and PFP_N (Di and Cameron, 2002; Ramirez-Garcia et al., 2014). In addition, drought reduces soil N mineralization and N availability, while N deficiency lowers the hydraulic conductivity and root water uptake efficiency (Wright et al., 2003; Austin et al., 2004). Such mutual dependencies highlight the importance of combined water-N management in achieving simultaneous improvements in WP, AE_N, and PFP_N. Studies show that precise N application reduces soil evaporation and improves water-holding capacity in mid-soil layers (Chen et al., 2010a; Lloret et al., 2010). In addition, optimized N regimes improve root architecture, with increased root length density and exploration, and thus increase water uptake from deep soil layers and increase WP (Lloret et al., 2010). Similarly, optimized irrigation scheduling enhances nitrogen remobilization from vegetative to reproductive organs, improving AE_N and PFP_N in grain crops (Lu et al., 2021; Zhao et al., 2024b). Thus, such practices reduce environmental hazards through decreasing N leaching and N_2O emissions while enhancing N assimilation in roots.

The physiological interaction between nitrogen and water provides a scientific basis for their integrated management in cropping systems (Sadras and Richards, 2014). Water is the main limiting factor for N assimilation processes, and N availability strongly regulates stomatal conductance and osmotic adjustment in the water metabolisms (Chapman et al., 2012; Ramirez-Garcia et al., 2014). This bidirectional interdependence governs the capacity for water and N acquisition, transport, and utilization in plants, suggesting that unilateral optimization of either factor leads to reductions in WP, AE_N, and PFP_N. Even though there has been a lot of research on how water and N work together, most of what we know comes from studying single crops (Sadras and Richards, 2014). There is still limited attention to intercropping and diversified cropping systems. This is important because intercropping and crop rotations are widely practiced globally. The lack of detailed studies on how different plant species compete and support each other for water and N in crop diversification makes it hard to use the benefits of water-N teamwork (Chen et al., 2022; Dong et al., 2024a). Filling this research gap could lead to substantial improvements in agricultural sustainability, particularly in water-limited regions where effective management of water and N is critical for stable yields and positive environmental outcomes (Zhang et al., 2015; Wang et al., 2022a). Future research should prioritize investigating how soil microbial communities interact with climate change in diversified cropping systems, to fully explore the potential of integrated water and N management strategies. Moreover, different intercropped species often have asynchronous water and N demands, significantly increasing the operational complexity of field management (Chen et al., 2022). Future research and extension efforts must explicitly define economic thresholds and physiological constraints, ensuring that water-N strategies are adapted to the financial capacity and specific soil-climate conditions of local farming systems.

4.3. Optimizing tillage and mulching techniques

Effective water and N management relies not only on the precise application of inputs but also on the soil's ability to retain them. These soil properties can be enhanced by rational tillage and mulching practices, which are essential components of sustainable intensification by improving root activity, soil microbial diversity, and biogeochemical cycling processes (Houx et al., 2011; Lyu et al., 2025). Such management practices create favorable rhizosphere conditions in which crop growth dynamics are aligned with the timing of water and N availability, resulting in substantial increases in WP, AE_N, and PFP_N (Masvaya et al., 2017; Silva et al., 2020). Recent studies show that conservation tillage systems, especially no-till with straw retention, can increase soil water infiltration rates by over 15% in wheat||maize intercropping systems compared with conventional tillage in oasis irrigation areas, while at the same time lowering soil evaporation via canopy shading and residue cover at the soil surface (Baumhardt et al., 2013; Yin et al., 2020a). Plastic film mulching has pronounced microclimatic effects, enhancing deep soil water use through vapor barrier effects and temperature regulation (Wu et al., 2017; Wan et al., 2019). Nevertheless, the advantages can be offset due to secondary effects such as increased soil temperature, which commonly becomes 3 −5°C higher than that of uncovered soil, and thus can advance root senescence and shorten crop growth duration (Bu et al., 2013; Yin et al., 2020a), leading to reduced yield and lower WP, AE_N, and PFP_N. The thermal stress caused by the warming effect of plastic mulching can also affect soil microbial communities, especially nitrifying bacteria, and create complex feedback effects on N mineralization rates.

Vertical nutrient stratification in soils with varied tillage practices represents a challenge and opportunity for precision management (Houx et al., 2011; Varnagirytė-Kabašinskienė and Survila, 2025). Typical no-tillage and intercropping maize increased the accumulation of available N by 60 −70% in the upper 10 cm soil layer such that a nutrient-rich band supports early establishment and morphogenesis but potentially constrains late-season crop growth owing to subsoil nutrient deficiencies in North America (Bossolani et al., 2020). Traditional deep-plowing at a depth of 30 −40 cm can break up hard-setting plow pans that inhibit root establishment, improving subsoil access for N in medium-textured soils (Silva et al., 2020). The integration of organic amendments with tillage strategies shows particular promise. Straw mulching in no-tillage systems may reduce topsoil temperatures by 2 −3°C during periods of high air temperature periods in arid northwest China, straw incorporation with 15 −20 cm in the soil can maintain thermal stability while improving N retention through enhanced microbial immobilization (Yin et al., 2023; Song et al., 2024). When practiced together with intercropping systems with complementary root architectures, such as combined practices can maximize synergistic effects and result in over 20% increase in WP in maize with wheat and pea intercropping compared to monoculture in northwest China arid irrigation areas (Yin et al., 2015; Zhao et al., 2019). No-tillage can increase AE_N and PFP_N by more than 13% when applied in wheat-maize rotation systems in arid irrigation regions of northwest China (Yang et al., 2020; Tan et al., 2022). Regional climates must strictly constrain the application of tillage and mulching techniques. While plastic film mulching is highly effective in cool, arid environments for conserving moisture and raising soil temperature during early growth stages, its application in warmer regions or during high-temperature seasons incurs high agronomic and environmental costs (Bu et al., 2013; Wan et al., 2019). However, current understanding remains largely empirical and experience-based. Key knowledge gaps exist in understanding both the physiological processes of root-shoot signaling under diversified tillage-mulching practices and the regulation of soil microbial community structure and function in relation to water and N use (Liu et al., 2024a; Lyu et al., 2024). Future research should prioritize elucidating the physiological, biochemical, and biological mechanisms underpinning water and N efficiency in diversified cropping systems under

different tillage and mulching regimes.

4.4. Establishing intelligent monitoring and decision support

The convergence of Unmanned Aerial Vehicles (UAVs), Internet of Things (IoT)-based sensor networks, and machine learning (ML) technologies is gradually transitioning water and N management in diversified crop systems. Currently, in terms of existing application foundations, UAVs with multispectral and thermal sensors have demonstrated mature capabilities in high-definition crop and soil health monitoring, allowing for precision irrigation and fertilization (Vega et al., 2015). In-field IoT sensors supplement UAV data with real-time soil moisture and N measurements, enabling more precise resource management (García et al., 2020). Building upon this foundation, developing technologies are moving towards predictive and generative management. ML models are increasingly being tested to process such datasets for the purpose of determining optimal crop arrangements, forecasting resource needs, decreasing interspecies competition, and vastly enhancing WP, AE_N, and PFP_N (Chlingaryan et al., 2018). For example, reinforcement learning models are currently being explored to model intercropped systems in order to determine arrangements which maximize resource complementarity while reducing environmental footprints (Thapa et al., 2018; Klompenburg et al., 2020), and this contributes to of WP, AE_N, and PFP_N.

Scalability of such technologies requires a multipronged effort involving the role of the farmer, policy support, and technological innovations. Modular decision-support tools can transform ML advice into implementable maps for planting, while interoperable tools can automatically roll out and can lead to precise management along with efficient N and water use (Kamilaris et al., 2017). Policymakers should incentivize the uptake of UAV-based services, shared-data platforms, and capacity-building programs among smallholders (Khanna et al., 2024). Hybrid AI approaches that integrate mechanistic crop models with interpretable ML techniques can improve user trust among farmers by providing more transparent and interpretable outputs(Jones et al., 2017). Multi-scale implementation depends on iterative feedback loops in which farmer-generated data are used to refine algorithms, ensuring that recommendations are better aligned with local site conditions (Jones et al., 2017). These developments can support sustainable precision agriculture by enhances productivity, resilience, and resource use efficiency across diverse agroecosystems (Kamilaris et al., 2017). Despite these technological advances, the widespread adoption of monitoring and decision-support systems remains constrained by clear socioeconomic and infrastructural limitations. Key barriers high capital costs of equipment, the need for specialized technical expertise, and limited data connectivity in remote rural areas, all of which disproportionately affect smallholder farmers (Khanna et al., 2024). Furthermore, the performance of ML models depends strongly on the quality and representativeness of local training data; applying generic models in uncalibrated or highly heterogeneous environments may result in inaccurate agronomic recommendations and inefficient resource use (Chlingaryan et al., 2018; Klompenburg et al., 2020). Therefore, before large-scale deployment, these systems should be critically evaluated against local economic conditions, infrastructure capacity, and specific management requirements to avoid over-reliance on technologies that may not yet be fully adapted to local contexts.

5. Current challenges and future research requirements regarding WP, AEN, and PFPN in diversified crop cultivation

5.1. Current challenges regarding WP, AE_N, and PFP_N in diversified crop cultivation

5.1.1. The complexity of selecting suitable variety combinations

Crop diversification is an important direction for improving WP, AE_N, and PFP_N of crops. Selecting the optimal combination of crops remains highly challenging in practice. This is due to several different factors that interact with each other, such as mutual competition among multiple species, compensatory utilization of resources in combinations, and variations in ecological niches with time and space. We must also consider the genetic variations among the crops themselves. Currently, selection of crop combinations relies largely on short-term field trials, which can consider only a few combinations and fail to capture long-term effects under changing agricultural and climatic conditions. In addition, there is no comprehensive database linking crop characteristics with their performance in diversified systems, making it difficult to assess and make informed decisions. This knowledge gap assumes greater significance now, as climate change plays an increasingly important role and constrains the extent to which the strategy can be implemented on a scale.

5.1.2. Obstacles to mechanized operation and large-scale application

The shift from monoculture systems to diversified production systems poses large-scale challenges with machinery, and so the potential for enhancing water and N use efficiency at scale remains limited. Traditional agricultural machinery is mainly designed for the uniform planting, fertilization, and harvesting of a single crop, and is difficult to adapt to the complexity of the spatial and temporal configuration in diversified systems. Especially in intercropping patterns, there is a lack of specialized machinery that can coordinate the needs of different crops, leading farmers to rely heavily on manual operations, thereby increasing production costs. Precision resource management is also limited. Most of the existing variable irrigation and fertilization technologies are unable to distinguish the different water and N requirements of diversified cropping systems, resulting in excessive water and N inputs and decreased utilization efficiency. Moreover, in rotation systems, the integration of cover crops or green manures connections often conflict with the fixed cycles of mechanized operations, affecting the continuity of the operations. If these mechanization obstacles are not systematically overcome, diversified cropping systems as a sustainable path for improving WP, AE_N, and PFP_N will be severely constrained in industrialized agricultural areas with labor shortages.

5.1.3. The assessment methods for the comprehensive benefits of economy, ecology, and society are inadequate

Currently, the comprehensive assessment of the economic, ecological and social benefits of diversified crop planting systems remains a core challenge in promoting their application to enhance WP, AE_N, and PFP_N. Existing assessment methods often fail to fully capture their multidimensional impacts, resulting in undervaluation or misjudgment of their value. Economic appraisals of cropping systems often focus on short-term payback, overlooking long-term system resilience and capacity for climate risk mitigation. Ecological assessments, meanwhile, typically exclude field-scale indicators, making it difficult to accurately quantify the effects of varied crop associations on soil moisture, N dynamics, and biodiversity. The social dimension is even more neglected, largely because factors such as labor allocation, gender, and local knowledge are difficult to quantify systematically quantification, despite their critical role in shaping the feasibility and equity of implementation. The resulting lack of comprehensive assessment creates a negative spiral, where weak evidence leads to weak policy support, causing farmers to proceed cautiously with necessary transformations. Ecological advantages, such as the enhancement of soil health, also take years to become visible, whereas assessments are often based on single growing seasons, creating a temporal mismatch. Interactions among multiple benefits also lack complete model-based exploration and so result in fragmented understanding and inhibit decisions based on evidence for sustainable transformation in agriculture.

5.2. Future research focuses on the efficient utilization of water and N in crop diversification

5.2.1. Research on the adaptability of diversified planting systems under climate change scenarios

Future research in diversified planting systems should focus on adaptation to climate change, and effort should be specifically directed toward the improvement of WP, AE_N, and PFP_N in highly fluctuating environmental conditions. Research should prioritize the identification of 'climate-smart' crop combinations that sustain resource use efficiency under suboptimal conditions. To this end, integrated process models, encompassing crop physiology, soil ecology, and microclimate dynamics are needed to simulate and predict system behavior under fluctuating climatic scenarios. New technologies should be employed to understand the dynamic behavior of complex systems and to generate actionable insights that are accessible to farmers. In combination with high-throughput machine learning-based phenotyping systems, it should become feasible to identify the crucial functional traits with potentiality for WP, AE_N, and PFP_N improvement in defined climatic conditions. The latter will provide the background for breeding in diversified planting systems and further investigation toward climatic adaptability in diversification of crops. In addition, more broad-based system resilience indexes should become available. Those indexes should carry the potential for WP, AE_N, and PFP_N in resisting climatic shocks, and be included in an evaluation matrix probing their interactions with yield stability and ecosystem services.

5.2.2. Analysis of the interaction mechanism of diversified cultivated crops at the molecular level

Future studies in crop diversification should aim at unraveling the molecular-level interactions in diverse crop combinations in order to optimize WP, AE_N, and PFP_N. A multi-omics approach with the incorporation of genomics, transcriptomics, proteomics, and metabolomics should be employed to holistically investigate how crop assemblies regulate the expression of gene sets associated with water transport and enzymes involved in nitrogen metabolism. This contributes to unveiling the synergistic processes at the molecular level that enable enhanced resource capture, by examining the role of root exudates, signaling molecules, and nutrient exchange pathways in mediating interspecies communication. It has the potential to transcend the limitations inherent in older experience-based selection practices, and to integrate reductionist approaches in molecular biology with systems-level agronomy research. This integrated approach can guide the design of optimal diversified cropping systems. Such systems should be based not only on phenotypic performance criteria but also on a molecular-level understanding of synergistic interactions among crop species. Such an approach should provide the scientific basis for sustainable use of crop systems with respect to water and N management.

5.2.3. Systematic analysis of the water-carbon-nitrogen coupling process

Crop diversification can significantly enhance WP, AE_N, and PFP_N, benefit soil health, and help address climate change. Nevertheless, the mechanisms through which the processes among water, carbon, and nitrogen cycles interact in this system remain poorly understood. To comprehensively understand these processes, models, remote sensing, and machine learning technologies must be combined and used in long-term observation to examine how multiple crop combinations increase WP, AE_N, and PFP_N by managing interactions among water, carbon, and N. Some key research directions are: (1) Quantifying how soil moisture, N dynamics, and carbon stocks respond to different crop combinations; (2) Investigating the exchange processes of water, carbon, and N among plants, soil, and the atmosphere with stable isotopic ($\delta^{15}N$, $\delta^{13}C$) and eddy covariance technologies, and examining the trade-offs between yield stability and WP, AE_N, and PFP_N; (3) Developing an accurate decision support system for irrigation and fertilization with the aid of real-time information in respect of water, carbon, and N. Understanding the interactions among water, carbon, and N in diversified cropping systems will be beneficial in developing key insights in designing sustainable crop production systems in the context of climate-smart agriculture.

5.2.4. Design of a precise and diversified planting system assisted by artificial intelligence

Traditional farming systems depend on basic irrigation and fertilizer advice, which does not account for intricate interactions among the water, N, and the variety of crops. We can analyze soil moisture and N dynamics, climate features, and the growth stages of the crop with the aid of artificial intelligence techniques such as machine learning and deep neural networks (Fig. 4). We can thus develop site-specific management plans. Furthermore, AI crop selection models can suggest desirable blends of rotation and intercropping based on how multiple crops complement one another in utilizing the water and N. The key research directions are: (1) Developing precise digital representations (digital twins), referred to as the digital twins, of the farms where the technologies such as the IoT, drones, and satellites are integrated to enable precise monitoring of water and N; (2) Developing transparent AI models and training them to interpret ecological interactions in the field and provide valuable advice to the agronomists; (3) Utilizing edge computing to provide real-time assistance in resource allocation. Addressing these challenges would enable AI-driven precision and diversified agriculture to enhance crop yields while improving water and nitrogen use efficiency. Such advances are crucial in the context of global change, as they support both environmental sustainability and long-term agricultural growth.

5.2.5. Establishing diversified planting systems suited to modern agricultural development

The establishment of diversified planting should align with green, intensive, and sustainable agricultural technologies globally. Specific crop types or combinations of varieties should be developed and selected for such systems, and mechanization technologies should be optimized. This should include upgrading green production techniques for suppression of pests, pathogens, and weeds, while promoting the adoption of fully mechanized farming systems. These approaches will enhance the productivity of diversified planting systems while ensuring sustainable use and maintenance of farmlands. The ultimate goal is to establish a high-quality, efficient diversified planting systems characterized by the seamless integration of machinery and agronomic practices, low resource input, environmental friendliness, and consistently high and stable yields.

5.2.6. The applicability and promotion strategies of diversified planting systems

Diversified planting systems are widely distributed and highly heterogeneous. It is necessary to strengthen their role in promoting green agricultural development based on regional characteristics. From the aspects of top-level design, policy guidance, and ideological emphasis, fully recognize the importance of diversified planting in improving the quality of cultivated land, strengthening ecological functions, ensuring food security, and increasing WP, AE_N, and PFP_N. Increase policy and financial support in regional macro agricultural planning, planting structure adjustment, and agricultural scientific and technological innovation. In regions with favorable conditions, research efforts should accelerate the development and demonstration of diversified cropping systems. This includes establishing region-specific, multi-functional pilot systems and formulating comprehensive technical packages and extension-application strategies that cover the entire production process.

6. Conclusions

Crop diversification is a key strategy for addressing the growing challenge of limited arable land alongside increasing global food

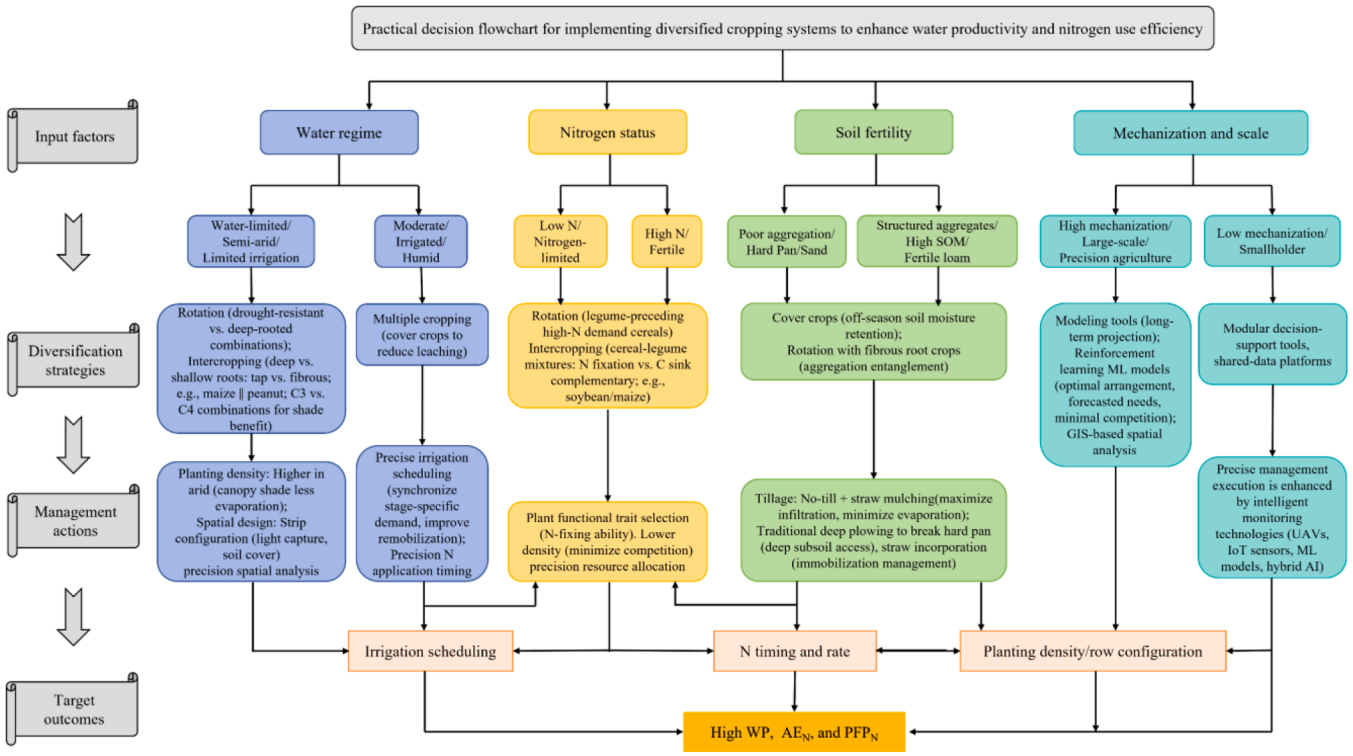


Fig. 4. Practical decision flowchart for implementing diversified cropping systems to optimize water productivity and nitrogen use efficiency. This top-down hierarchical framework guides the selection of context-specific diversification strategies and precision management actions based on primary environmental and operational inputs. Input factors encompass water regime, nitrogen status, soil fertility (including structural integrity and soil organic matter), and mechanization scale. These contexts inform the optimal diversification strategies, such as specific crop rotations, intercropping patterns (e.g., deep/shallow root complementarity, legume-cereal mixtures), and multiple cropping sequences. To actualize these strategies and management actions, including precision irrigation scheduling, targeted N timing and rate, and optimized planting density or spatial configuration, must be synergistically implemented and dynamically supported by intelligent monitoring technologies. The integration of these components ultimately targets the maximization of system-level resource efficiencies. WP, Water productivity; AE_N, Agronomic efficiency of nitrogen; PFP_N, Partial factor productivity of nitrogen; SOM, Soil organic matter; ML, Machine learning; UAVs, Unmanned aerial vehicles; IoT, Internet of things; GIS, Geographic information system.

demand. It can enhance yield per unit area while improving WP, AE_N, and PFP_N. This review shows that diversified cropping systems improve these efficiencies by reducing resource losses, enhancing soil properties, regulating interspecific interactions, and promoting beneficial microbial activity. Effective implementation requires optimized crop configurations, improved water and nitrogen management, appropriate tillage and mulching practices, and the use of monitoring systems. However, the performance of these systems and associated interventions (e.g., plastic mulching, spatial configurations) depends strongly on management practices, local climate conditions, and baseline soil fertility. Inappropriate configurations or poor aligned tillage practices may lead to increased resource competition, premature root senescence, and severe yield penalties. Despite these advances, challenges remain in variety selection, mechanization, and the integrated assessment of economic, ecological, and social outcomes. Future research should focus on improving climate adaptability, understanding the mechanisms underlying water and nitrogen use, and advancing integrated water–carbon–nitrogen interactions, supported by data- driven and precision management approaches. To support practical implementation, policymakers should promote diversified cropping systems through targeted subsidies, technical extension services, and investment in smart farming infrastructure. Overall, integrating crop diversification with optimized agronomic management can improve resource use efficiency, support ecosystem functions, and contribute to sustainable crop production.

CRediT authorship contribution statement

Lianhao Zhao: Resources, Data curation. **Falong Hu:** Resources, Data curation. **Zhilong Fan:** Resources, Data curation. **Cai Zhao:** Project administration, Data curation. **Aizhong Yu:** Project administration, Data curation. **Pingxing Wan:** Resources, Data curation. **Yunyou Nan:** Methodology, Investigation. **Pete Smith:** Writing – review & editing, Methodology, Investigation. **Wen Yin:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Qiang Chai:** Writing – review & editing, Funding acquisition, Formal analysis, Conceptualization. **Heyu Chen:** Writing – review & editing, Data curation, Conceptualization. **Mohamed Abdalla:** Writing – review & editing, Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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Glossary

AE_N: Agronomic efficiency of nitrogen
AI: Artificial intelligence
AMF: Arbuscular mycorrhizal fungi
CMN: Common mycorrhizal network
CNKI: China national knowledge infrastructure
IoT: Internet of things
ML: Machine learning
MWD: Mean weight diameter
PFP_N: Partial factor productivity of nitrogen
SOC: Soil organic carbon
SOM: Soil organic matter
UAV: Unmanned aerial vehicles
WP: Water productivity